

AKSELOYA, Svalbard and Jan Mayen

WMO: 010170

Lat: 77.689N

Lon: 14.784E

Elev: 20

StdP: 14.69

Time Zone: 1.00 (EUC)

Period: 10-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	-12.1	-7.0	N/A	N/A	N/A	N/A	N/A	N/A	42.3	20.4	39.2	22.5	10.5	60	1.001

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.9	49.5	N/A	47.9	N/A	46.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	13.2	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 37.2	32.8	28.9	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.1	19.4	18.1	13.7	18.3	29.0	37.7	42.7	42.1	38.3	31.0	24.9	20.8		
	DBStd	12.67	11.60	12.88	11.50	9.05	5.26	2.86	2.45	3.10	4.22	6.45	7.99	9.22		
	HDD50	8007	950	893	1124	950	652	368	226	245	351	588	754	906		
	HDD65	13482	1415	1313	1589	1400	1117	818	691	710	801	1053	1204	1371		
	CDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	13.5	16.4	16.4	14.7	13.1	12.5	9.5	10.2	10.3	12.2	13.3	16.0	17.3		
	PrecAvg	20.5	2.3	1.9	1.9	1.2	0.9	0.7	1.2	1.6	2.2	2.0	2.3	2.2		
	PrecMax	28.6	6.0	4.3	4.4	3.2	2.0	1.7	2.6	3.7	4.3	4.1	7.5	4.5		
(15) Precipitation	PrecMin	12.7	0.4	0.4	0.6	0.4	0.3	0.2	0.2	0.6	0.6	0.7	0.6	0.6		
	PrecStd	4.1	1.5	1.1	1.0	0.6	0.4	0.4	0.6	0.6	0.9	0.9	1.3	1.0		
	DB	41.2	40.7	38.8	38.0	41.9	47.2	51.5	53.2	49.8	45.7	42.0	42.0	42.0		
	MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2% DB	36.8	38.1	36.1	34.7	38.7	44.9	49.1	50.1	47.8	43.5	39.6	36.6	36.6		
(22) Mean Coincident Wet Bulb Temperatures	MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% DB	34.6	36.0	33.3	32.9	36.8	43.2	47.5	48.2	45.7	42.0	37.7	34.4	34.4		
	MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10% DB	32.1	32.9	29.5	30.5	35.4	41.7	46.2	46.4	43.8	39.8	35.0	32.3	32.3		
	MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2% WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(32) Mean Daily Temperature Range	MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10% WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% DB	6.9	7.8	7.9	7.6	5.4	4.4	4.9	4.8	4.8	4.8	6.0	6.7	6.7		
	MCDBR	8.9	9.1	10.2	8.0	6.4	6.4	7.0	6.9	7.6	7.3	7.7	8.3	8.3		
(36) Clear-Sky Solar Irradiance	MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	taub	N/A	0.196	0.256	0.286	0.303	0.296	0.302	0.291	0.249	N/A	N/A	N/A	N/A		
(40) All-Sky Solar Radiation	taud	N/A	1.906	2.084	2.228	2.268	2.384	2.447	2.447	2.369	N/A	N/A	N/A	N/A		
	Ebn at Noon	0	16	179	245	262	272	265	247	200	N/A	0	0	0		
	Edh at Noon	0	2	16	25	30	28	25	21	13	N/A	0	0	0		
(44) All-Sky Solar Radiation	RadAvg	0	9	243	843	1390	1669	1363	886	336	42	0	0	0		
	RadStd	0	1	28	108	106	120	110	79	33	4	0	0	0		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)		+2.74	+8.33	+7.38	N/A	N/A	N/A	-994	-988	N/A	N/A	N/A

Nomenclature: See separate page